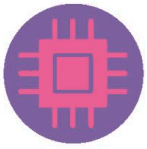
Digital integrated circuits

Integrated circuits (ICs), also known as "chips," typically contain billions of tiny transistors. Each one is either on or off (allowing current through or not), representing the binary numbers 1 and 0. Combinations of these numbers represent the letters, images, and sounds that make up the files on a computer, as well as the programs that make computers work.

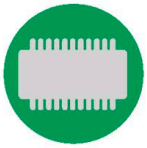
CAN TRANSISTORS KEEP GETTING SMALLER?

Chip designers are close to the limit of how small silicon transistors can be, but with new materials, such as compound semiconductors, they will shrink them even further.

Types of digital integrated circuit
Digital integrated circuits are designed to do specific jobs. Electronics engineers fit them together with other components on a circuit board to make digital devices such as computers, tablets, smartphones, and digital cameras.



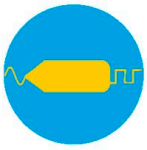
Microprocessor
Every digital device has an IC that processes programs—sets of instructions that make the device work.



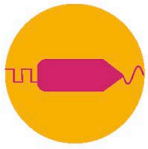
RAM chip
Random access memory (RAM) holds active programs and information to be processed.



Graphics chip
Graphics chips send signals to the screen of a computer, smartphone, or tablet, rapidly refreshing the display.



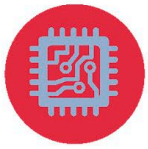
Analogue to digital
An analogue-to-digital chip takes information from the real world and codes it into collections of binary numbers.



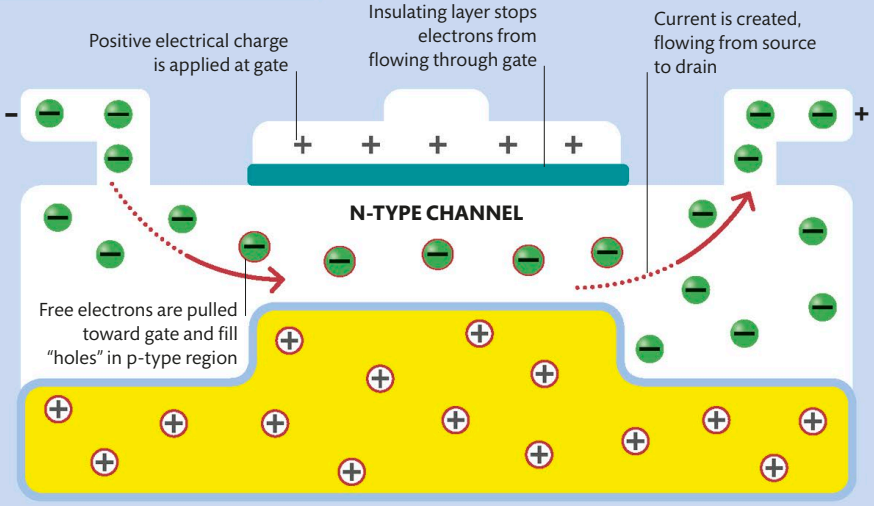
Digital to analogue
A digital-to-analogue chip processes digital sound (1s and 0s) to produce a signal that can be sent to loudspeakers.

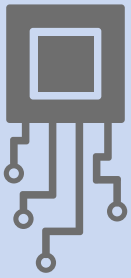


Flash memory chip
Found in USB storage, digital cameras, and solid-state hard drives, flash chips can store huge amounts of information.



System on a chip
An IC that contains all the circuits found on most other types of IC can be used as a stand-alone computer.





EACH **TRANSISTOR** IN A MEMORY CHIP **STORES 1 SINGLE "BIT"**

Transistor is "on"
A positive charge at the gate attracts negatively charged electrons into the p-type region. These can become charge carriers that allow current to flow through the transistor.